

Utilizing Video and Sensor Technology for Analyzing Non-Swimmer Children in Swimming Education Programs through Observational Methods

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ABSTRACT

Accidental drowning stands as the third leading cause of global mortality, according to the World Health Organization. While suggestions like increased supervision and physical barriers aim to reduce drowning risk, their effectiveness is hindered by the widespread accessibility of water bodies. In contrast, swimming education has demonstrated a remarkable potential to decrease the risk of drowning by up to 90%. However, research on technology-supported swimming education is limited, mainly focusing on drowning detection and support for competitive swimmers. This paper addresses this gap by presenting outcomes from a user study, particularly focusing on a vulnerable demographic: children without basic swimming skills in survival-oriented beginner courses. Data from 45 participants aged 4 to 13 were collected through wearable sensors and cameras. The study's results offer valuable insights into integrating technology into swimming education for non-swimming children, contributing to the advancement of drowning prevention strategies.

CCS CONCEPTS

• **Applied computing** → *Computer-assisted instruction*; • **Human-centered computing** → *User studies*.

KEYWORDS

Swimming Education, Wearable Technology, User Studies

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1 INTRODUCTION

Drowning, as reported by the World Health Organization, stands as the third leading cause of global fatalities, posing a particularly grave threat to children under 13, who are five times more susceptible to accidental drowning [17, 20, 23]. The primary factors attributed to this alarming statistic include access to dangerous

water [18], lack of supervision [16], and a lack of swimming proficiency [1]. According to a study by Brenner et al., individuals possessing swimming and water safety knowledge are 90% more likely to survive drowning incidents [1].

Swimming education primarily involves two approaches: parental teaching and professional instruction. Parental teaching allows for a more relaxed and personalized learning experience, fostering a sense of safety for the child. However, this method relies on the parent's swimming abilities, and with approximately 50% of Americans unable to swim, many children miss out on this opportunity [5]. Consequently, parents often turn to swim instructors, who offer expert training in survival swimming techniques that parents may not possess, enhancing a child's chances of surviving a drowning incident. Nevertheless, the group setting in which instructors typically teach can limit personalized attention and potentially slow down a child's progress in learning to swim. Therefore, there is a growing need to explore technology-driven solutions that can bridge the gap between these two teaching methods.

Combining the strengths of both parental teaching and professional instruction through technology could offer a promising solution. Such a hybrid approach could provide personalized swimming lessons while leveraging the expertise of professional instructors. It would ensure that children receive comprehensive swimming education, including essential survival skills, regardless of their parents' swimming abilities. This innovative approach could help children become proficient swimmers more efficiently, addressing the current limitations of both traditional teaching methods and contributing to improved water safety for future generations.

While technological advancements have been applied to various aspects of swimming, the existing research predominantly revolves around two specific contexts. The first centers on lap swimming, a domain mostly inhabited by seasoned swimmers or experts. Studies in this realm predominantly focus on utilizing activity recognition to identify the four primary swimming strokes (backstroke, breaststroke, butterfly, and freestyle) [4, 24]. The second context pertains to drowning incidents, where video-based activity recognition attempts to detect signs of drowning, such as erratic movements or unconsciousness while floating in or above water [11, 13]. However, both of these scenarios fall short when it comes to addressing swimming education and rectifying the drowning problem. In lap swimming, the participants are already proficient swimmers, leaving little room for improving technique or performance. In drowning incidents, the challenges lie in defining sporadic movements and addressing the potential medical complications that arise

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when an individual becomes unconscious, even if they ultimately survive [21].

In light of these limitations, our study focuses on the integration of technology within a swimming education environment. Our participants had minimal to no prior swimming experience, and we collected data from wearable sensors placed on their lower back and wrist, as well as video footage. This paper aims to address three pivotal research questions:

- What kind of environment is conducive for conducting user studies with individuals who lack swimming skills and are learning to swim?
- What activities do swimming education programs teach?
- What insights can be gleaned from integrating technology into a swimming education curriculum?

Recognizing the crucial importance of swimming education in preventing drownings, there is a pressing need to offer a more accessible and efficient method for individuals to learn how to swim. In partnership with a swimming school in Austin, we have gathered video and sensor data from children engaged in their swimming program. Our research aims to illuminate the essential steps in collecting, analyzing, and observing participants in potentially perilous situations, with the ultimate aim of improving swimming education through the integration of technology.

2 RELATED WORK

In recent years, technological advancements have led to the emergence of waterproof and water-resistant devices, such as Fitbit, Garmin watches, and Apple Watches. These innovations have not only transformed the way we interact with technology in aquatic environments but have also given rise to a burgeoning field of aquatic activity research. In this section, we delve into three distinct categories of activities that have witnessed significant integration of technology: swimming, swimming education, and drowning prevention. These domains represent the most prevalent areas where technology is harnessed to either enhance performance or safeguard lives. Our investigation has revealed a predominant focus on activity recognition, a feature notably embedded in commercial devices like Fitbit and Apple Watches. Within this section, we provide insights into the types of user studies conducted, the target populations involved, and the spectrum of activities recognized by these technologies. Additionally, we explore the current state-of-the-art technologies employed to support these three activity domains. Our primary aim is to illuminate the current trajectory of technological influence in aquatic activities and underscore the significance of this paper, which specifically addresses the role of technology in swimming education, particularly among novice swimmers.

2.1 Swimming Activity Recognition

Swimming activity recognition has predominantly dominated research efforts when compared to studies on drowning and educational aspects. This dominance can be attributed to the availability of a readily accessible population for study. The majority of previous research papers in this domain have focused on college swimming teams engaged in competitive training, often preparing for events such as the Olympics [6, 9]. Consequently, these studies primarily revolved around lap swimming, involving the execution of various

swimming techniques (backstroke, breaststroke, butterfly, freestyle) and returning.

In contrast, our research takes a distinctive approach by concentrating on a more vulnerable population—children with little to no prior swimming experience who are endeavoring to acquire swimming skills. Our research setting also differs significantly, as we operate within a classroom environment where participants are eager to learn how to swim. It's worth noting that, while Chen et al. did include children in their study, these children were already part of a swim team and proficient in swimming techniques and form [3]. Our focus, on the other hand, is on collecting data from individuals in the early stages of their swimming journey, which involves assessing beginner-level swimming forms.

Regarding data collection, previous studies have employed various methods, including video data and sensor data [8, 24]. Recent years have witnessed a shift towards the collection of sensor data, often from wrist-worn devices [24], which can be attributed to the availability of commercial devices for this purpose. Our user study emphasis lies on collecting data from two primary areas: the wrist and the lower back. These areas are chosen for their significance in swimming activity recognition and their classification as non-intrusive, ensuring that participants are not subjected to discomfort during data collection.

2.2 Drowning Activity Recognition

Research in the field of drowning has presented considerable challenges owing to the elusive nature of the activity itself. Defining drowning has proven to be a complex task, with various perspectives in academic literature. Some studies perceive drowning as attempts to ascend a metaphorical ladder or grasp at any available support on land, while others characterize it as marked by irregular or sporadic movements [19]. This inherent difficulty in defining drowning has led some researchers to adopt an approach of recognizing everything except drowning or focusing on common drowning scenarios, such as individuals floating headfirst or submerged underwater [12]. However, conducting empirical studies to validate these hypotheses is inherently perilous, as simulating real drowning scenarios poses significant risks to human subjects.

To circumvent these dangers, some researchers have turned to video data as the predominant medium for integrating technology into drowning and activity recognition research. For instance, Handalage et al. have concentrated their efforts on distinguishing between four distinct activities: treading water, swimming, floating, and splashing around to address irregularity in movement classification [10]. Nonetheless, it is crucial to acknowledge the inherent limitations of these approaches, as they rely on the assumption that floating necessarily indicates a critical stage of drowning, where the individual is unconscious and in imminent danger of severe harm due to oxygen deprivation from water ingestion.

In contrast, our research aims to prioritize swimming education as a proactive strategy to mitigate the risk of drowning. Our approach focuses on teaching individuals proper swimming techniques to prevent them from entering a drowning situation and equipping them with the skills to survive such situations if they do arise. This educational approach has been substantiated by previous studies, which have demonstrated a remarkable 90% reduction

in drowning incidents among participants who received proper swimming education [1].

2.3 Swimming Education Activity Recognition

Swimming education encompasses a wide range of topics, including coaching and swim team dynamics, as well as structured courses or classroom settings designed to prevent drownings through teaching individuals how to swim. Within the realm of technology-based research, the predominant focus has been on supporting swim coaches by providing them with data such as stroke counts, lap times, and overall motion analysis [7, 22]. A relevant study by Moren et al. delves into working with beginners by affixing sensors to individuals' ankles, even while they are clothed, to evaluate their swimming capabilities [14]. This study revealed that the presence of sensors did hinder swimming performance but had no significant impact on buoyancy.

In our research, we embark on a novel exploration within the field of swimming education, particularly in teaching individuals with very limited to no swimming abilities. We conduct an observational study that examines participant interactions with wearable technology and assesses the study's performance by analyzing data collected from cameras and sensors. Our objective is to provide valuable insights for those interested in conducting similar studies, offering guidance on the types of activities to record, the best practices for data collection, and the lessons we have learned from this study, along with its implications and outcomes.

3 SYSTEM OVERVIEW

The primary objective of this study was to incorporate technology into the realm of swimming education and to investigate the process of how individuals acquire swimming skills, all while amassing valuable data about the learning journey. Prior research efforts have focused on gathering camera and sensor data from individuals using wearable devices. In our study, we adopted similar methodologies and obtained camera and wearable device data. However, we recognized the unique challenges presented by the aquatic environment, which necessitated a comprehensive exploration of available options. Consequently, we also gathered sensor data from participants' wrists and lower back, as these areas are minimally invasive and benefit from the increasing availability of wrist-based smart devices. This section will delve into the various devices employed, the critical considerations when utilizing them, and the nature of the data collected. Furthermore, we will outline the software tools used for data analysis, highlighting how we extracted valuable insights from the device data.

3.1 Wearable Devices

For our data collection, we aimed to gather information from wearable sensors and camera recordings. Regarding wearable data collection, our focus was on the wrist and lower back areas. To accomplish this, we sought to employ a commercially available device. However, our choice of device was contingent on its suitability for use in water, as water resistance was crucial for our project. We evaluated devices using their Ingress Protection (IP) rating, which assesses their waterproof capabilities, ranging from 0 to 8.

Our goal was to find a device capable of functioning even when submerged to depths of up to 10 meters. We settled on this depth because it represents the deepest part of a pool, and given our involvement in swimming education, we needed equipment that could endure underwater conditions. It's worth noting that many devices can only handle water up to a meter deep, with only a select few capable of withstanding greater depths. Notably, the Apple Watch Series 2 and Polar M600 Smartwatch impressed us with their ability to function even at depths of up to 50 feet underwater.



Figure 1: Data collection device and wearable strap for lower back

For the wrist area, we opted for the sleek and compact Apple Watch, primarily due to its compatibility with children of various sizes. Our Institutional Review Board (IRB) stipulated an age limit of 4+, necessitating a small watch for smaller wrists. As for the lower back, we selected the Moto 360. Figure 1 illustrates our solution: an elastic wearable strap designed to accommodate the Moto 360, with added Velcro for adjustable sizing. The strap can comfortably fit waist sizes ranging from 20 to 42 inches, which aligns with the average measurements of 4 to 18 year-olds. Simplicity was key in our design, with a basic satchel—a sewn pocket in the middle of the belt—offering ample storage. Given our young participants' limited swimming experience and physical exertion, we aimed for an unobtrusive design capable of withstanding less force from water.

In terms of data collection, we encountered the challenge of working with two different operating systems, namely Apple and Android. Nevertheless, we successfully developed software that could seamlessly collect data from both devices. Notably, both devices were capable of producing sensor data at a rate of 100 Hz. The sensor data we gathered encompassed accelerometer readings, heart rate data (from the wrist), gyroscope data, and barometer data—these being the most commonly collected sensor data in previous research papers. The overall data was stored in a CSV with timestamps in milliseconds and each column associated with each sensor.

3.2 Camera Devices

When selecting the camera for our project, our primary considerations were affordability and minimal disruption to the swimmers. We wanted a solution that could be easily repositioned without causing any inconvenience. After careful deliberation, we decided to utilize a mobile device, specifically the Nexus 6 mobile phone.

This choice was driven by the impressive advancements in mobile technology, enabling it to capture video at an optimal rate of 100 frames per second.

Our swimming lessons typically lasted for 30 minutes, and during our initial testing, we found that recording video in 4K resolution at 100 frames per second quickly consumed the device's storage within just 5 minutes. To overcome this challenge, we made adjustments by reducing the video resolution to 2K and lowering the frame rate to 30 frames per second. This optimization allowed us to collect 30 minutes of valuable data while utilizing only 2 gigabytes of storage, well within the capacity of the 10-gigabyte memory available on the device.

With the camera parameters resolved, our next task was to find an appropriate mounting solution that would elevate the device sufficiently to capture the entire pool area. For this purpose, we opted for selfie stick tripods that could extend up to 6 feet in height, exceeding our requirements. These tripods were not only adjustable but also collapsible, affording us the flexibility to use multiple tripods for various research studies.

3.3 Data Analysis Software

Once the data has been collected, it requires analysis and observation. The dataset comprises CSV files containing sensor data from wristwear and video data. To synchronize and annotate this data, a system capable of extracting video segments, annotating them with timestamps and labels was necessary. Although various systems were available, many incurred costs or demanded a steep learning curve. Our requirement was simplicity: a tool that could efficiently mark time intervals with labels based on specified start and stop times.

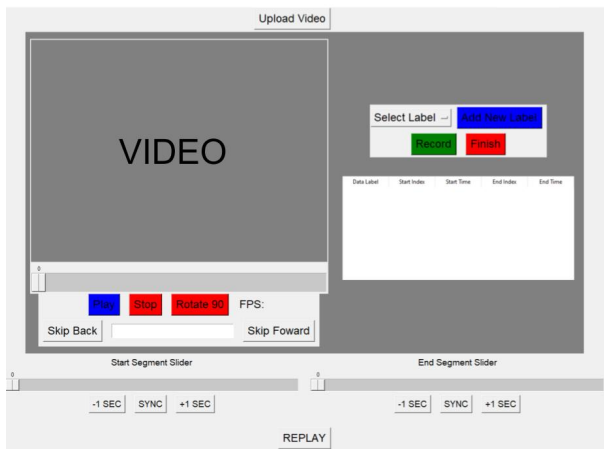


Figure 2: User interface of video labeling system

In Figure 2, we present our developed system designed to process CSV and video files. This system incorporates a skip feature, which operates based on video index, and an adjustable frames-per-second (fps) setting. Additionally, users have the capability to add custom labels and record the labeled video data, complete with start and stop times and corresponding indices, in CSV format. The overall system is user-friendly and serves its labeling purpose effectively.

The labeling categories are defined by common activities typically occurring during class, as elaborated upon in the methodology section. Furthermore, we have included a "BAD" label category to identify video segments with suboptimal recording quality, such as instances where someone obstructs the camera's view or when the camera experiences disruptions, among other potential issues.

4 METHODOLOGY

4.1 Research Design

Our objective is to observe and gather participant data during their swimming learning process. To achieve this, we designed a study that enables us to interact with participants and collect both visual and sensor data. Our approach involved conducting an ethnographic study where we closely observed participants during their swimming lessons while simultaneously capturing sensor data. To effectively execute this study, it was crucial for us to gain a comprehensive understanding of how swimming schools operate.

In pursuit of suitable partners for our research, we reached out to several swimming schools. After engaging in discussions with them, we established a collaboration with Texas Swimming School, which was willing to work with us. The Texas Swimming School employs a unique approach to teaching beginners to swim, organized into ten distinct stages. Each stage focuses on teaching a specific swimming stroke, with each stroke building upon the skills acquired in previous stages.

It was crucial to identify the specific swimming activities for our study. Unlike traditional swimming, which involves four primary strokes, survival swimming (or competency swimming) lacks a universally defined set of activities. However, previous research papers have described survival swimming as encompassing floating, turning from front to back and vice versa, propulsion swimming (freestyle, backstroke, breaststroke, sidestroke), and treading water [2, 15]. The Texas Swimming School addresses question 2 by offering ten stages, covering activities such as floating, swimming, entering/exiting the water, and rotating oneself in the water. These ten stages include most of the essential activities that need to be taught, as outlined in the list below:

- (1) Holding Breath Underwater - Able to hold breath and put head underwater
- (2) Float on Back - Float stationary on back in the water
- (3) Float on front - floating stationary on front in the water
- (4) Kick off wall - kick or push off wall while in water
- (5) Kick floating on back - kicking while floating on the back for propulsion
- (6) Freestyle - arms and leg propulsion with person on the stomach
- (7) Backstroke - arms and leg propulsion with person on back
- (8) Roll over back to front - rotating body from back to front while in water
- (9) Roll over front to back - rotating body from front to back while in water
- (10) Jump in and recover to wall - jumping into water from land and able to swim back to top and breach water

In our research, we propose gathering data from participants within a single class session. Our intention is not to mandate the use

of sensors for multiple classes, given that most children usually attend one class per week. The particular level or stage of instruction for each child's participation is still undecided, contingent upon their individual preferences and readiness to engage in our study. This uncertainty arises from our approach of collecting data from willing participants, making it challenging to pinpoint a specific stage for focus.

4.2 Data Collection

We aimed to gather swimming data from both the wrist and lower back using video recording to document the activities and their corresponding timestamps. As we observed that swimming classes followed a staged progression, it became evident that children practiced by traversing the pool from one side to the other.

To facilitate data collection, we devised a procedure wherein we activated the sensors on the lower back before inviting the child to participate in our study. Since these sensors were integrated into a strap, it was more convenient to activate them in advance, as waiting for the child to put it on could be challenging.

Regarding the video recording, once the child had the devices in place and resumed their swimming classes, we assigned a researcher to reposition the camera and relocate to the opposite side of the pool. This particular side was less crowded and provided a clear view, as it was frequented not only by instructors but also by external monitors ensuring the class proceeded smoothly.

Upon completion of the child's swimming session, we retrieved the devices and categorized them according to the camera associated with each one. Subsequently, we halted the data recording applications, collecting data from accelerometers, gyroscopes, barometers, and video recordings.

4.3 User Study Procedures

In our user study, we enrolled participants aged four and above. This age range was selected based on discussions with the local swim team, as they indicated that most of their students were four years old or older. Our consent procedure involved a two-step approach. Initially, we engaged with parents using a spoken script to recruit those interested in having their children participate in the study. After these conversations, parents would sign a consent document. Once approval was obtained, a researcher would approach the child, either while they were waiting for their class to begin or potentially during the class itself, and present them with a verbal script. For the child, a verbal agreement was all that was required to proceed with the study.

The task assigned to the child involved wearing a wearable weight device along with a lower back strap device while attending their class. Once they had completed this task, they would return the equipment to the researcher to confirm its completion. The final step involved the researcher signing an assent document, affirming that they had received permission from the child.

Overall, the procedures were successful, with some initial hesitancy from parents. However, communicating with the children proved to be easier, as the devices appeared engaging and enjoyable. It is important to note that when placing the devices on the children, they should be at least 3 feet away from water, as they are a vulnerable group without swimming skills. While we encountered no

issues, safety remained a top priority, and we wanted to emphasize this point.

4.4 Data Analysis

After gathering all the data, we faced the challenge of lacking labeled data, and the video data lacked synchronization. To address this, we employed two methods. The first involved labeling the data based on the child's swimming activities, which was facilitated by our pre-existing activity list. This list allowed us to associate each activity with the appropriate label. Additionally, we added a few activities of our own, such as entering the water, holding on to the wall, sitting, standing, exiting the water, and instructor interaction.

Instructor interaction emerged as the most crucial activity, as it involved physical engagement between the instructor and the child, guiding them in improving their swimming technique. This interaction encompassed actions like holding the child and assisting with floating or guiding their arm movements while learning the backstroke. Another significant activity we noted was sitting, as it highlighted the need to recognize that each child received individual instruction. Instructors often had multiple children to teach simultaneously, necessitating others to sit for safety, a surprising but important aspect.

The system we developed was user-friendly, with minimal usability issues. The primary challenge was synchronizing the start and stop points of each activity, but once this was mastered, the overall video analysis became much more efficient. Nevertheless, the workload on researchers was demanding, as they had to focus on the videos and label more than 30 activities per video, which could be draining. However, the system itself was straightforward, and with the predefined list of activities, labeling them became a manageable task.

4.5 Ethical Considerations

We need to recognize that we are dealing with a vulnerable population, specifically children who are not proficient swimmers. This necessitated addressing several ethical concerns.

First and foremost, constant supervision was imperative. As a requirement of the swimming school, parents were required to be present in a separate room with windows, ensuring that they had visual access to everything transpiring in the pool area. Additionally, our instructors maintained vigilant oversight both in the water and around the pool.

Another critical consideration was the implementation of safety measures to prevent any potential drowning or harm during the study. We took great care to ensure that the watch and lower back devices were non-intrusive and elastic, permitting unrestricted movement. These devices were designed to be lightweight, eliminating any possibility of impeding the children during their practice sessions. Moreover, we placed the devices on the children at a safe distance of 3 feet from the pool to minimize the risk of accidental slips or falls. Finally, before the researcher's departure, they ensured that the child was seated precisely where they had disembarked to facilitate ongoing supervision by the instructors.

5 RESULTS AND DISCUSSION

In our user study, data was gathered from 45 participants (ages ranging from 4 - 13), comprising approximately 30 minutes of video footage and wearable sensor data (Wrist/Lowerback) from each participant. The researchers made key observations regarding the participants' interactions with the wearable devices and the conduct of the classes, based on the video data and on-site observations. This section will delve into a quantitative analysis of the swimming classes, as well as the researchers' findings concerning interactions between participants, other individuals, and the on-site devices.

5.1 Q1 Answer - Observations Analysis

We had to anticipate what to expect while conducting the user study in this unfamiliar environment, and we had concerns about the willingness of both children and parents to participate. Additionally, we needed to plan for how to execute the user study effectively. The swimming school was divided into two groups: Morning and Afternoon sessions. Morning classes typically had no more than ten children, whereas the afternoon sessions were significantly larger, accommodating around 30 to 50 participants, nearly triple the size. The school's meticulous organization, which kept track of each student's progress, proved invaluable.

Collecting data during the morning sessions was much easier compared to the afternoon. This was due to more students in the afternoon, which increased the frequent passage of parents and children in front of the camera during the transition between classes. However, this was only an issue when the class we were recording ended. To optimize data collection in the future, we recommend installing a high-up camera system, preferably on a wall or approximately 10 feet above the ground.

The school's structure included swimming instructors and two pool managers. Instructors were responsible for teaching at each skill level, with approximately ten instructors present in the pool. The pool managers' role was to ensure the safety of all students and guide them to their appropriate skill level. What was visually notable is that even during swimming lessons instructors and managers constantly communicated. The reason is that the managers approved when a child has learned a skill also they provided insight to the lesson at a higher elevation which helped the instructors.

The school's excellent organization was crucial, especially since many parents were initially hesitant to allow their children to participate in the study. After obtaining parental approval, researchers had to locate the child within the large group. Collaboration with the swimming school staff, particularly the pool managers, facilitated this process. Once the child was identified, they were briefly removed from their class to seek their consent for participation. Upon approval, the child was equipped with the wearable device, and the study began.

Interestingly, while parents were initially hesitant, the children themselves were enthusiastic about participating. They were eager to wear the wearable watch and interact with it throughout the study. One particular incident underscored this enthusiasm: when one child agreed to participate, their older sibling in the next class became jealous and attempted to acquire a device of their own.

A notable observation was that children tended to be more distracted by the wearable watch compared to a device located on their

lower back. They frequently touched and interacted with the watch. Fortunately, the device was locked, preventing accidental shutdown or app deactivation. However, this heightened interaction posed a challenge, as it made it challenging for children to stay engaged while waiting for their turn to swim.

Furthermore, a recurring practice among the children was waiting, sitting by the pool, and then swimming from one side to the other. This typically involved practicing water entry, swimming, and then exiting the pool. Some children had to sit on the pool wall, which was a safety measure because they lacked strong swimming skills. Thankfully, the facility provided a warm and comfortable environment for them.

Lastly, we observed that regardless of their skill level, the swimming school had all students jump into the water from the poolside and swim to the wall. This was a unique and intriguing teaching method that we had not anticipated but now appreciate as an important aspect of the swimming curriculum.

5.2 Q3 Answer - Quantitative Analysis

To answer Question 3, we need to understand how the participants swim through quantitative analysis. As mentioned previously, we gathered video and sensor data from a total of 45 participants. In our quantitative analysis, we will delve into the activities undertaken, including the quantity and duration of each activity. Additionally, we will explore the interactions of participants, encompassing those who engaged in swimming, interacted with instructors, and spent time on the pool wall.

During each class session, a child spends approximately 30 minutes, with an additional 5 minutes allocated for administrative tasks like sign-in and transitioning to the appropriate stage. In some cases, this transition period may exceed the designated time frame. We noticed that each instructor could accommodate 1 to 4 students per class. The number of students per class in each study were as follows: Study 1 had 4 students, Study 2 had 8 students, Study 3 had 27 students, and Study 4 had 6 students. This data illustrates that Study 3 had the highest number of classes with 3 students per class. On the other hand, Studies that contained 1, 2, and 4 students had significantly fewer classes. This difference in students per class can be observed in Figure 3 with 3 having less instructor time than 4.

Our observations revealed that children spent more time sitting on the pool wall than actively swimming. Figure 3 illustrates that when an instructor oversees two or fewer children, the child is more likely to be engaged in swimming activities compared to scenarios where there are three or more children per instructor. This can be attributed to the fact that the child swims across the pool with the instructor closely supervising, requiring the instructor to traverse the pool to assist the next child. Consequently, the instructor divides their time evenly between children on either side of the pool. Figure 3 further demonstrates that when children spend less time waiting and sitting on the wall, their interactions with instructors become more substantial. This is primarily due to instructors having fewer children to attend to, enabling them to allocate more individualized time and attention. The need for students to interact with their peers also extends the practice duration, contributing to this observation.

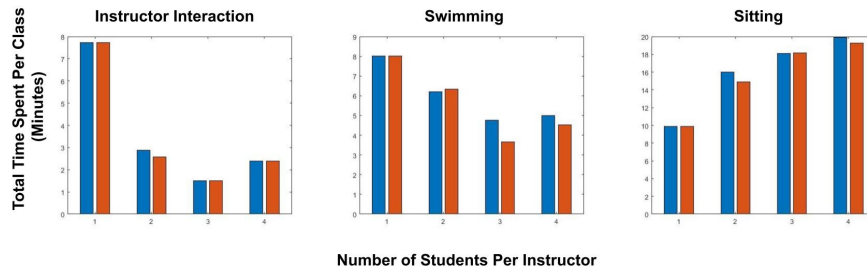


Figure 3: Quantitative analysis of time spent for each major activity for the participant during class (BLUE - Median, RED - Mean)

6 CONCLUSION

Drowning, the third leading cause of global fatalities, can be effectively prevented through various strategies, with acquiring swimming skills and water safety knowledge emerging as the most impactful. While existing research has predominantly focused on professional swimmers or the critical stages of drowning, our investigation takes a unique approach. Centered on a vulnerable demographic – children undergoing swimming lessons – our user study reveals a surprising pattern. Children often spend minimal time actively swimming, primarily remaining seated on the pool wall with brief intervals of swimming practice. This study not only sheds light on the challenges faced by this population but also demonstrates how technological studies can be conducted to address them. By sharing our insights, we aspire to guide and inspire others interested in similar studies, contributing to the enhancement of swimming activities for children learning how to swim.

7 FUTURE WORK

We aim to leverage the sensor and video data we’ve gathered from individuals in the process of learning how to swim. Our goal is to delve deeper into these activities, extracting essential features that will be instrumental in the creation of machine learning algorithms designed for activity classification. Once our system is developed, our intention is to seamlessly integrate it into swimming education. Additionally, we plan to explore innovative methods of supporting both instructors and swimmers throughout their learning journey. Given that swimmers spend a significant portion of their time waiting, we’re considering the integration of gamification and technology interaction to engage and assist them during these idle moments.

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